

In the sections that follow, we will look at the configuration of a FreeBSD system and its processes. We will learn about configuration through `rc.conf`, and the startup scripts `/usr/local/etc/rc.d`. Plus, we will learn about network cards and how their configuration and testing work. We will also learn a lot about network devices and the configuration of virtual hosts on them. The chapter will also teach us about the usage of configuration files included in `/etc`. In the tuning section of this chapter, we will learn about the tuning of the operating system through `sysctl` variables, disk performance tuning, and the modification of kernel limitations.

As with any such esoteric and specialised operating system as FreeBSD, there are a few prerequisites to reading this chapter. You must be well acquainted with the basics of FreeBSD as well as UNIX. You must also know about the configuration as well as the compilation of kernels to enable you to better reproduce the code and concepts included in this chapter.

With that out of the way, let's get to understanding tuning and configuration in FreeBSD!

Starting services

Third-party software in FreeBSD is generally installed from the Ports collection. Users install third-party software and then configure them to start during initialisation.

Through `rc.d`, the configuration of third-party apps and services is only a few lines of code. In FreeBSD, you can also set up certain third-party services to be initialised on the initialisation of another third-party service, instead of having all services initialised during the startup phase. Instead of the flags coded into the startup script, they can be passed through `/etc/rc.conf`.

An example of this is the following code. The code develops a way to set the process ID and track it. It also requires the DAEMON pseudo service to start before the startup of the required utility.

CODE:

- `#!/bin/sh`
- `#`
- `# PROVIDE: utility`
- `# REQUIRE: DAEMON`
- `# KEYWORD: shutdown`